Ivan Lau | Curriculum Vitae

National University of Singapore, School of Computing
https://ivanphlau.github.io/ ivan.lau@u.nus.edu

CURRENT RESEARCH INTERESTS

Sequential decision-making under uncertainty and/or communication constraints

EDUCATION

Ph.D. in Computer Science August 2024 - Present
National University of Singapore

Research topics: Sequential Learning Problems with 1-bit Feedback
CGPA: 5.0/5.0
Advisor: Jonathan Scarlett

Graduate student in Electrical and Computer Engineering August 2022 - December 2023
Rice University

Research topics: Decentralized Optimization
CGPA: 4.00/4.00
Advisors: César A. Uribe and Shiqian Ma

M.Sc. in Mathematics September 2019 - August 2021
Simon Fraser University

CGPA: 4.07/4.33
Thesis: Nonuniform Compressed Sensing Schemes with Sublinear Measurements, Sublinear Time, and Low Entropy (awarded Certificate with Distinction)
Advisor: Jonathan Jedwab

B.Sc. (Hons) in Computer Science and Mathematics September 2015 - June 2019
University of Edinburgh

Degree classification: First-Class Honours
Mathematics project: Left Braces and the Solutions of the Yang-Baxter Equation (awarded best project)
Informatics project: Hermitian Spectral Theory of Mixed Graphs

EMPLOYMENT

National University of Singapore January 2024 - August 2024
Research Assistant, Department of Computer Science

Rice University May 2023 - December 2023
Research Assistant, Department of Electrical and Computer Engineering

National University of Singapore September 2021 - August 2022
Research Assistant, Department of Computer Science

Simon Fraser University September 2019 - August 2021
Research Assistant, Department of Mathematics

University of Edinburgh September 2017 - May 2019
Teaching Assistant, School of Informatics

University of Edinburgh June 2018 - August 2018
Research Intern, Laboratory for Foundations of Computer Science

PUBLICATIONS

Journal Papers

8. **Model-Based and Graph-Based Priors for Group Testing**
   Ivan Lau, Jonathan Scarlett, Yang Sun
   IEEE Transactions on Signal Processing, 2022.

7. **An Associative Left Brace is a Ring**
   Ivan Lau
   Journal of Algebra and Its Applications, 2020

Machine Learning Conference Papers

6. **Open Problem: Is Interaction Necessary for Order-Optimal 1-bit Mean Estimation?**
   Ivan Lau, Jonathan Scarlett
   Conference on Learning Theory (COLT) 2026.

5. **Sequential 1-bit Mean Estimation with Near-Optimal Sample Complexity**
   Ivan Lau, Jonathan Scarlett
   International Conference on Artificial Intelligence and Statistics (AISTATS) 2026.

4. **Quantile Multi-Armed Bandits with 1-bit Feedback**
   Ivan Lau, Jonathan Scarlett
   International Conference on Algorithmic Learning Theory (ALT), 2025.

3. **Max-Quantile Grouped Infinite-Arm Bandits**
   Ivan Lau, Yan Hao Ling, Mayank Shrivastava, Jonathan Scarlett
   International Conference on Algorithmic Learning Theory (ALT), 2023

IEEE Conference Papers

2. **Decentralized and Equitable Optimal Transport**
   Ivan Lau, Shiqian Ma, César A. Uribe
   American Control Conference (ACC), 2024

1. **Construction of Binary Matrices for Near-optimal Compressed Sensing**
   Ivan Lau and Jonathan Jedwab
   IEEE International Symposium on Information Theory (ISIT), 2021.

PREPRINTS

In Submission

2. **Batched Stochastic Linear Bandits with 1-Bit Communication Constraints**
   Ivan Lau, Daniel McMorrow, Kevin Jamieson, Jonathan Scarlett

1. **Order-Optimal Sequential 1-Bit Mean Estimation in General Tail Regimes**
   Ivan Lau, Jonathan Scarlett

TEACHING EXPERIENCE

National University of Singapore

The Algorithm Designer's Toolkit (Teaching Assistant)	Spring 2025

Rice University

Random Signals (Teaching Assistant)	Fall 2023

Simon Fraser University

Calculus Workshop (Teaching Assistant)	Fall 2020
Applied Calculus Workshop (Teaching Assistant)	Summer 2020
Computing with Linear Algebra (Teaching Assistant)	Spring 2020
Algebra Workshop (Teaching Assistant)	Spring 2020, Fall 2019

University of Edinburgh

Informatics 2B - Algorithms, Data Structures, Learning (Teaching Assistant)	Spring 2019
Informatics 2D - Reasoning and Agents (Teaching Assistant)	Spring 2019, Spring 2018
Algorithms and Data Structures (Teaching Assistant)	Spring 2019
Discrete Mathematics and Mathematical Reasoning (Teaching Assistant)	Fall 2018, Fall 2017
Informatics 1 - Cognitive Science (Teaching Assistant)	Spring 2018

AWARDS AND HONOURS

Malaysian National Mathematical Olympiad (4th - 13th Place Bracket)	2013
Malaysian Computing Competition (Perfect Score)	2014, 2013
University of Edinburgh William and Isabella Dick Fourth Year Project Prize	2019
National University of Singapore Research Scholarship	2024
Rice University Department of Electrical and Computer Engineering Fellowship	2022
Simon Fraser University Department of Mathematics Certificate "With Distinction"	2021
Simon Fraser University Graduate Fellowship and Special Graduate Entrance Scholarship	2019
International Competitions and Assessments for School (Gold Medal in Science)	2012

PROFESSIONAL ACTIVITIES

Reviewer for IEEE Trans. Inf. Theory	2026 - Present
Reviewer for AISTATS, ICLR, NeurIPS	2025 - Present
Reviewer for ICASSP, ACC	2024 - Present

Skills

Python, R, MATLAB, LaTeX